DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular Winter Examination – 2024

Course: B. Tech. **Branch: Common to all branches** **Semester: I**

Subject Code & Name: (24AF1000BS101) Engineering Mathematics - I

Max Marks: 60 **Date: 06/02/2025** **Duration: 3 Hr.**

Instructions to the Students:

1. *Each question carries 12 marks.*
2. *Question No. 1 will be compulsory and include objective-type questions.*
3. *Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.*
4. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
5. *Use of non-programmable scientific calculators is allowed.*
6. *Assume suitable data wherever necessary and mention it clearly.*

					(CO)	Marks
Q.1	**Objective type questions. (Compulsory Question)**					**12**
1	Homogeneous system of linear equations is/has				(CO1)	1
	a. always consistent	b. always inconsistent	c. no solution	d. None		
2	The rank of matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix}$ is equal to				(CO1)	1
	a. 1	b. 2	c. 3	d. None		
3	If $A = [a_{ij}]$ is a square matrix of order n, then trace of matrix A is				(CO1)	1
	a. product of diagonal elements	b. sum of diagonal elements	c. sum of row elements	d. None		
4	If $u = f\left(\frac{y}{x}\right)$ then $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ is equal to				(CO2)	1
	a. 2u	b. u	c. 0	d. None		
5	If $u = x^y$ then $\frac{\partial u}{\partial y}$ is equal to				(CO2)	1
	a. $x^y \log x$	b. 0	c. yx^{y-1}	d. None		
6	If $u = f(y-z, z-x, x-y)$ show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}$				(CO3)	1
	a. 1	b. **0**	c. -1	d. None		
7	The condition for the function $f(x, y)$ to have maximum value at (a, b) is				(CO3)	1
	a. $rt - s^2 \geq 0$ *and* $r < 0$	b. $rt - s^2 > 0$ *and* $r > 0$	c. $rt - s^2 < 0$ *and* $r < 0$	d. None		
8	If $x = uv, y = \frac{u}{v}$ then $\frac{\partial(x,y)}{\partial(u,v)}$ is				(CO3)	1
	a. $\frac{-2u}{v}$	b. $\frac{-2v}{u}$	c. 0	d. None		
9	The formula for $\int_0^{\frac{\pi}{2}} sin^n\theta d\theta$ is equal to				(CO4)	1
	a. $\frac{n(n-1)(n-3)\ldots\ldots}{n(n-2)(n-4)\ldots\ldots} \times \frac{\pi}{2}$	b. $\frac{n(n-1)(n-3)\ldots\ldots}{n(n-2)(n-4)\ldots\ldots} \times 1$	c. $\frac{n(n-1)(n-3)\ldots\ldots}{n(n-2)(n-4)\ldots\ldots} \times 1$ *or* $\frac{\pi}{2}$	d. None		

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10	The number of loops in the polar curve $r = asin2\theta$ are				(CO4)	1
	a. 4	b. 2	c. 6	d. None		
11	The value of $\int_0^2 \int_1^y xy\, dxdy$ is equal to				(CO5)	1
	a. 0	b. 1	c.-1	d. None		
12	In polar co-ordinate system (r, θ) : value of $dy\, dx$ is equal to				(CO5)	1
	a. $dr\, d\theta$	b. $rdr\, d\theta$	c. $r^2 dr\, d\theta$	d. None		
Q.2	**Solve the following.**					12
A)	Solve the equations: $x + 3y + 2z = 0;\ 2x - y + 3z = 0;\ 3x - 5y + 4z = 0$.				(CO1)	6
B)	Find the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} 1 & 0 & -4 \\ 0 & 5 & 4 \\ -4 & 4 & 3 \end{bmatrix}$				(CO1)	6
Q.3	**Solve the following.**					12
A)	If $x^x y^y z^z = c$, show that at $x = y = z$, $\frac{\partial^2 z}{\partial x \partial y} = -(x \log ex)^{-1}$.				(CO2)	6
B)	If z is a homogeneous function of degree n in x, y, then prove that $x^2 \frac{\partial^2 z}{\partial x^2} + 2xy \frac{\partial^2 z}{\partial x \partial y} + y^2 \frac{\partial^2 z}{\partial y^2} = n(n-1)z$.				(CO2)	6
Q.4	**Solve any TWO of the following.**					12
A)	Expand $f(x, y) = e^x \sin y$ in the powers of x and y as far as the terms of third degree.				(CO3)	6
B)	Test the function $f(x, y) = x^4 + y^4 - x^2 - y^2 + 1$ for maxima, minima and saddle point				(CO3)	6
C)	Find the maximum value of $x^m y^n z^p$ when $x + y + z = a$.				(CO3)	6
Q.5	**Solve any TWO of the following.**					12
A)	Evaluate $\int_0^\infty \frac{dx}{(1+x^2)^8}$.				(CO4)	6
B)	Trace the curve $x = a(\theta - \sin\theta),\ y = a(1 - \cos\theta)$ (Cycloid).				(CO4)	6
C)	Trace the curve $r = a \sin 3\theta$ (3 Leaved Rose).				(CO4)	6
Q.6	**Solve any TWO of the following.**					12
A)	Evaluate $\int_0^1 \int_0^x e^{x+y}\, dy\, dx$.				(CO5)	6
B)	Find the area of the circle $x^2 + y^2 = a^2$.				(CO5)	6
C)	Define and verify Cayley Hamilton theorem for the matrix $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & 1 & -1 \end{bmatrix}$				(CO2)	6
	***** End *****					

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